

12.159

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Question:

The determinant

$$\begin{vmatrix} 1+b & b & 1 \\ b & 1+b & 1 \\ 1 & 2b & 1 \end{vmatrix} \quad (0.1)$$

evaluates to

- a) 0
- b) $2b(b-1)$
- c) $2(1-b)(1+2b)$
- d) $3b(1+b)$

Solution:

Let the determinant of the given matrix be Δ . We will use row operations to simplify the matrix. Adding a multiple of one row to another does not change the determinant's value. This method is valid for $b \neq -1$.

$$\Delta = \begin{vmatrix} 1+b & b & 1 \\ b & 1+b & 1 \\ 1 & 2b & 1 \end{vmatrix} \quad (4.1)$$

First, we create zeros in the first column of the second and third rows using the first row as the pivot.

Apply the operation $R_2 \rightarrow R_2 - \frac{b}{1+b}R_1$:

$$\Delta = \begin{vmatrix} 1+b & b & 1 \\ b - \frac{b}{1+b}(1+b) & (1+b) - \frac{b}{1+b}(b) & 1 - \frac{b}{1+b}(1) \\ 1 & 2b & 1 \end{vmatrix} \quad (4.2)$$

$$= \begin{vmatrix} 1+b & b & 1 \\ 0 & \frac{(1+b)^2 - b^2}{1+b} & \frac{1+b-b}{1+b} \\ 1 & 2b & 1 \end{vmatrix} \quad (4.3)$$

$$= \begin{vmatrix} 1+b & b & 1 \\ 0 & \frac{1+2b}{1+b} & \frac{1}{1+b} \\ 1 & 2b & 1 \end{vmatrix} \quad (4.4)$$

Next, apply the operation $R_3 \rightarrow R_3 - \frac{1}{1+b}R_1$:

$$\Delta = \begin{vmatrix} 1+b & b & 1 \\ 0 & \frac{1+2b}{1+b} & \frac{1}{1+b} \\ 1 - \frac{1}{1+b}(1+b) & 2b - \frac{1}{1+b}(b) & 1 - \frac{1}{1+b}(1) \end{vmatrix} \quad (4.5)$$

$$= \begin{vmatrix} 1+b & b & 1 \\ 0 & \frac{1+2b}{1+b} & \frac{1}{1+b} \\ 0 & \frac{2b(1+b)-b}{1+b} & \frac{1+b-1}{1+b} \end{vmatrix} \quad (4.6)$$

$$= \begin{vmatrix} 1+b & b & 1 \\ 0 & \frac{1+2b}{1+b} & \frac{1}{1+b} \\ 0 & \frac{b+2b^2}{1+b} & \frac{b}{1+b} \end{vmatrix} \quad (4.7)$$

We can see that the third row is b times the second row ($R_3 = b \cdot R_2$). This indicates linear dependence. To make this explicit, we perform one final operation, $R_3 \rightarrow R_3 - bR_2$.

$$\Delta = \begin{vmatrix} 1+b & b & 1 \\ 0 & \frac{1+2b}{1+b} & \frac{1}{1+b} \\ 0 & \frac{b(1+2b)}{1+b} - b\left(\frac{1+2b}{1+b}\right) & \frac{b}{1+b} - b\left(\frac{1}{1+b}\right) \end{vmatrix} \quad (4.8)$$

$$= \begin{vmatrix} 1+b & b & 1 \\ 0 & \frac{1+2b}{1+b} & \frac{1}{1+b} \\ 0 & 0 & 0 \end{vmatrix} \quad (4.9)$$

Since the matrix has a row consisting entirely of zeros, its determinant is 0.

$$\Delta = 0 \quad (4.10)$$

The correct option is (a).